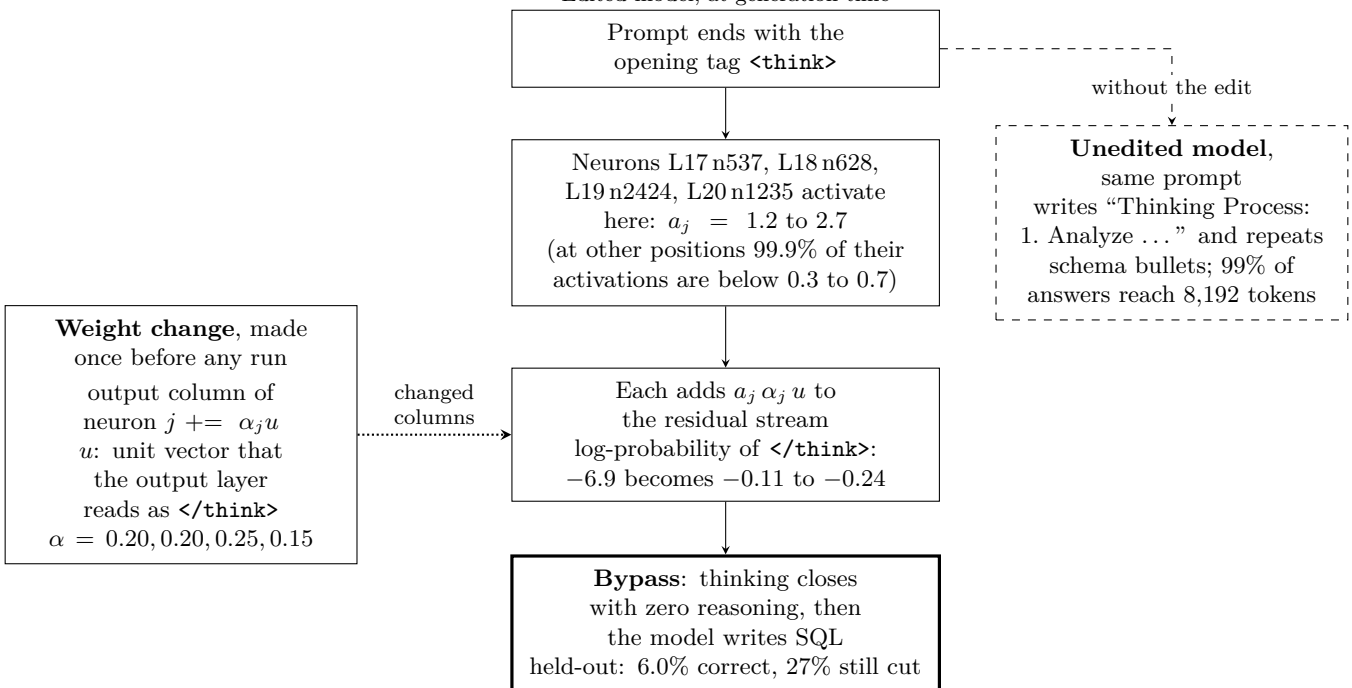
Edited model, at generation time
Prompt ends with the opening tag <think>
without the edit
Unedited model, same prompt writes "Thinking Process: 1. Analyze ..." and repeats schema bullets; 99% of answers reach 8,192 tokens
Neurons L17 n537, L18 n628, L19 n2424, L20 n1235 activate here: $a_j = 1.2$ to 2.7 (at other positions 99.9% of their activations are below 0.3 to 0.7)
Weight change, made once before any run
output column of neuron j += $\alpha_j u$
u: unit vector that the output layer reads as </think>
$\alpha = 0.20, 0.20, 0.25, 0.15$
changed columns
Each adds $a_j \alpha_j u$ to the residual stream
log-probability of </think>: −6.9 becomes −0.11 to −0.24
Bypass: thinking closes with zero reasoning, then the model writes SQL
held-out: 6.0% correct, 27% still cut